

Divyansh Shukla

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EDUCATION

Vellore Institute of Technology

B.Tech, Computer Science and Engineering (Artificial Intelligence & Machine Learning)

2023 – 2027

CGPA: 9.03 / 10

EXPERIENCE

Power Pulse India

2026 – Present

Machine Learning Intern

India

- Cut missed pedestrians 26% and raised mAP50-95 from 0.360 to 0.420 for a driver-assistance thermal detector, training a deployment-aware YOLOv8n on FLIR ADAS v2 across 6 road-safety classes in 100 epochs against 150, and trading 3 points of precision for the recall that safety case requires.
- Identified annotation inconsistency and input resolution as the primary accuracy bottlenecks through per-class crop audits and COCO size-bucket analysis: 82% of pedestrian labels fall under 32x32 px, and labelled trucks were delivery vans also annotated car and bus, with 37% of true trucks drawing no prediction at all.
- Ran a four-experiment resolution and architecture sweep to cut deployment cost, eliminating a P2 detection head that scored below plain 640px at 1.51x the compute and 4x the detection candidates, and identifying 800px as capturing 85% of the small-object recall gain at 1.56x compute against 960px's 2.25x.

CERTIFICATIONS

NPTEL: Applied Accelerated AI, Deep Learning with Computer Vision | **Coursera:** Deep Learning Specialization | **AWS:** AWS Cloud Architecting

TECHNICAL SKILLS

Languages: Python, SQL, C++, CUDA

ML Frameworks: PyTorch, PyTorch Lightning, Hugging Face Transformers, TRL, Unsloth, Ultralytics (YOLO), Stable-Baselines3, vLLM, llama.cpp

Deep Learning: flow/diffusion models, transformers, YOLO, generation, inpainting, detection, visual question answering (VLM), reinforcement learning (PPO)

Generative AI: LangChain, LangGraph, CrewAI, Google Agent Development Kit (ADK), Google Gen AI SDK, retrieval-augmented generation (RAG), EmbeddingGemma embeddings, pgvector, ChromaDB, agentic workflows (planning, memory, tool use), LLM evaluation

Cloud & Data: AWS (S3, EC2, Lambda, SageMaker, Redshift, EMR, ECS/ECR, IAM, CloudWatch), Google Cloud (BigQuery, Managed Service for Apache Spark / Dataproc, Dataflow, Pub/Sub, Cloud Storage, Vertex AI), Apache Spark, Modal

AI Developer Tools: Claude Code, GitHub Copilot, Gemini Antigravity, Gemini API

Model Deployment: ONNX, NCNN, INT8 quantization, embedded/edge inference

Simulation & Robotics: MuJoCo, Gymnasium, OpenCV, Raspberry Pi

Backend & Infrastructure: Docker, Kubernetes (minikube), FastAPI, Django, Flask, Weights & Biases, Hydra, Git, Linux, NumPy, pandas

PROJECTS

RoPE-DiT: Diffusion Transformer with 2D RoPE

github.com/divyanshshuklai/Custom2DRoPEDiT

- Developed a 5M-parameter Diffusion Transformer from scratch in PyTorch: patchifier, timestep and class-label embedders, AdaLN-Zero conditioning, and a custom multi-head attention module.
- Built a 2D rotary positional embedding that rotates queries and keys by each patch's height and width coordinates, splitting the head dimension into row and column halves so attention sees relative 2D distance.
- Trained on CIFAR-10 with a rectified flow objective and sampled with a hand-written Euler solver and classifier-free guidance.

Zero-Shot ICE RMA: Quadruped Locomotion RL

github.com/divyanshshuklai/zero-shot-ice-rma

- Built a MuJoCo environment for the Unitree Go1 quadruped with 30-dim proprioceptive observations, 12-dim joint-position actions, fractal terrain, and the Rapid Motor Adaptation bioenergetics reward with curriculum-scheduled penalties.
- Debugged a PPO baseline that ran 42M steps without learning a gait, tracing the exploding value loss to unnormalized observations and penalty dominance.

Fashion Search: Conversational Product Discovery

github.com/divyanshshuklai/fashion_search

- Built a FastAPI chat app answering natural-language product queries across 10 Indian e-commerce stores with Tavily search, Gemini 2.5 Flash extraction, and BeautifulSoup scraping.
- Wrote a priority-based image and price extractor that tries JSON-LD product schema, then Open Graph tags, then keyword-matched image tags, as the baseline for an agentic version with planning, memory, and tool use.

ACHIEVEMENTS

Won the internal (intra-campus) round of the Smart India Hackathon at Vellore Institute of Technology, 2024.